

$$\begin{aligned}
 0.3667 \varphi_2 - 0.2 \varphi_3 - 0.1 \varphi_4 &= 2.6667 \\
 -0.2 \varphi_2 + 0.4778 \varphi_3 - 0.1667 \varphi_4 &= -18.2222 \\
 -0.1 \varphi_2 - 0.1667 \varphi_3 + 0.3167 \varphi_4 &= 16.6667
 \end{aligned}$$

$T_{gs}$

$$G_3 = \frac{1}{R_3} = \frac{1}{15} = 0.066667 \text{ S}$$

$$G_2 = \frac{1}{R_2} = \frac{1}{9} = 0.111111 \text{ S}$$

$$G_5 = \frac{1}{R_5} = \frac{1}{20} = 0.05 \text{ S}$$

$$G_1 = \frac{1}{R_1} = \frac{1}{8} = 0.125 \text{ S}$$

$$G_6 = \frac{1}{R_6} = \frac{1}{10} = 0.1 \text{ S}$$

Решим систему с помощью матрицы

$$\varphi = (AG_DA^T)^{-1} (AJ_D - AG_DE_D)$$

После расчета получим

$$\varphi_2 = 10.551 \text{ V}$$

$$\varphi_3 = -17.394 \text{ V}$$

$$\varphi_4 = 46.809 \text{ V}$$

Определим токи ветвей

$$I_3 = \frac{\varphi_1 - \varphi_2 + E_3}{R_3} = \frac{0 - (10.551) + 100}{15} = 5.963 \text{ A}$$

$$I_2 = \frac{\varphi_3 - \varphi_1 + E_2}{R_2} = \frac{-17.394 - (0) + 50}{9} = 3.623 \text{ A}$$

$$I_5 = \frac{\varphi_4 - \varphi_1}{R_5} = \frac{46.809 - (0)}{20} = 2.34 \text{ A}$$

$$I_1 = \frac{\varphi_3 - \varphi_4 + E_1}{R_1} = \frac{-17.394 - (46.809) + 100}{8} = 5.966 \text{ A}$$

$$I_6 = \frac{\varphi_4 - \varphi_2}{R_6} = \frac{46.809 - (10.551)}{10} = 3.626 \text{ A}$$

$$I_0 = (\varphi_3 - \varphi_2) G_0 = (-17.394 - (10.551)) \cdot 0.2 = -5.539 \text{ A}$$

$$J_0 = J_0 = 4 \text{ A}$$